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Questions

Q1.

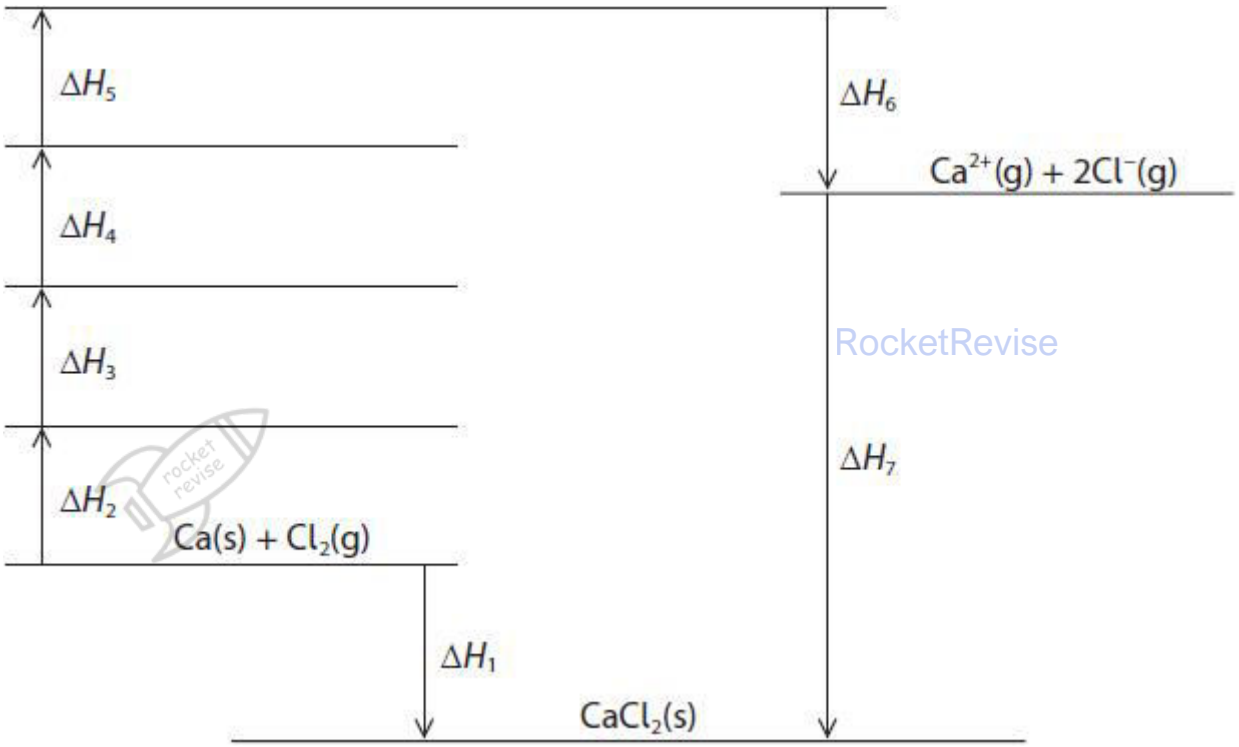
This question concerns calcium chloride, $CaCl_2$.

Some energy changes related to calcium chloride are shown.

Energy change	Value / kJ mol^{-1}
Atomisation energy of calcium	+178.2
Atomisation energy of chlorine	+121.7
Electron affinity of chlorine	–348.8
First ionisation energy of calcium	+590.0
Standard enthalpy change of formation of calcium chloride	–795.8
Experimental lattice energy of calcium chloride	–2 258
Second ionisation energy of calcium	To be calculated

(i) Complete the Born-Haber cycle by writing appropriate chemical formulae on the horizontal lines.
Include state symbols and relevant electrons.

(4)



(ii) Calculate the second ionisation energy of calcium using the cycle and the values given in the table.
Include a sign and units with your value.

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(3)

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(b) The theoretical lattice energy of formation of $CaCl_2$ is −2 223 kJ mol^{-1}.

Explain why the experimental lattice energy of $CaCl_2$ is more exothermic than the theoretical value.

(3)

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(c) The table shows the theoretical lattice energies of some chlorides of Group 2.

Group 2 metal chloride	magnesium chloride	calcium chloride	strontium chloride	barium chloride
Lattice energy / kJ mol^{-1}	−2326	−2223	−2127	−2033

Explain the trend in **theoretical** lattice energies on descending Group 2.

(3)

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(d) Calculate the enthalpy change of solution of calcium chloride, $CaCl_2$.

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Data

$\Delta_{hyd}H^{\ominus}(Ca^{2+}) = -1\,650\ kJ\ mol^{-1}$ $\Delta_{hyd}H^{\ominus}(Cl^{-}) = -364\ kJ\ mol^{-1}$

Experimental lattice energy ($CaCl_2$) = −2 258 kJ mol^{-1}

(2)

(Total for question = 15 marks)

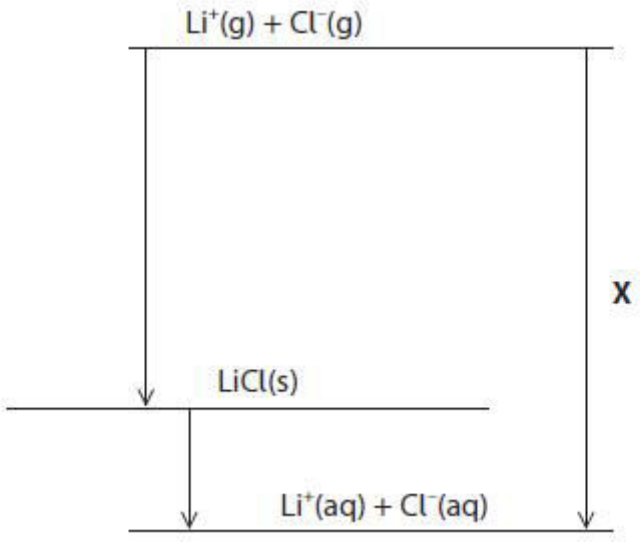
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Q2.

What does **X** represent on the diagram?

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- ☒ **A** sum of enthalpy changes of hydration of the gaseous ions
- ☒ **B** enthalpy change of formation of LiCl
- ☒ **C** enthalpy change of solution of LiCl
- ☒ **D** lattice energy of LiCl

(Total for question = 1 mark)

Q3.

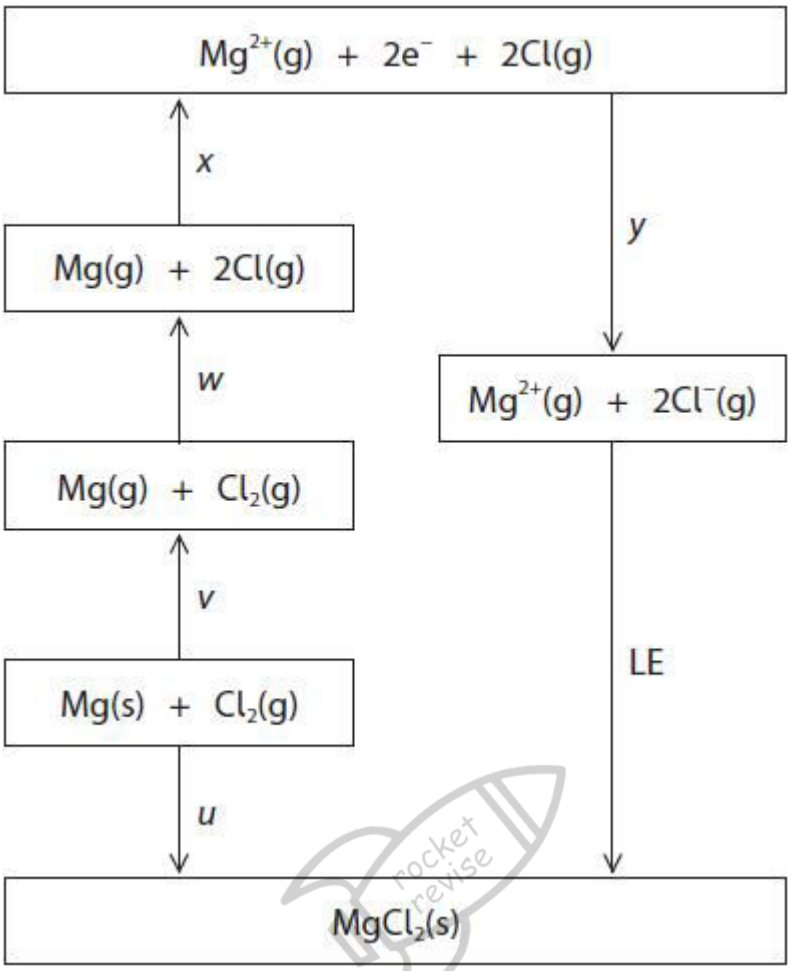
Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the Born-Haber cycle for magnesium chloride.

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(a) Which of these is the electron affinity of chlorine?

(1)

- **A** y
- **B** $y/2$
- **C** $(w + y)$
- **D** $(w + y)/2$

(b) Which expression gives the lattice energy (LE) for magnesium chloride?

(1)

- **A** LE = $u - (v + w + x + y)$
- **B** LE = $v + w + x + y - u$
- **C** LE = $y - u - (v + w + x)$
- **D** LE = $v + w + x - y + u$

(c) Which energy change in the cycle does x represent?

(1)

- **A** the first ionisation energy of magnesium
- **B** the second ionisation energy of magnesium
- **C** the sum of the first and second ionisation energies of magnesium
- **D** the sum of the enthalpy change of atomisation and the first and second ionisation energies of magnesium

(Total for question = 3 marks)

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Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which equation represents the standard enthalpy change of atomisation, $\Delta_{at}H$, of bromine?

☐ **A** $\frac{1}{2}Br_2(l) \rightarrow Br(g)$

☐ **B** $Br_2(l) \rightarrow 2Br(g)$

☐ **C** $\frac{1}{2}Br_2(g) \rightarrow Br(g)$

☐ **D** $Br_2(g) \rightarrow 2Br(g)$

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(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The enthalpy change of solution of sodium sulfate, Na_2SO_4 , may be calculated using three pieces of data. Which of these pieces of data is **not** required?

☐ **A** lattice energy of Na_2SO_4

☐ **B** enthalpy change of hydration of Na^+

☐ **C** enthalpy change of formation of Na_2SO_4

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☐ **D** enthalpy change of hydration of SO_4^{2-}

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about four ionic compounds.

(a) Which of these compounds would be expected to have the **least** exothermic lattice energy?

(1)

- ☐ **A** calcium chloride
- ☐ **B** magnesium chloride
- ☐ **C** potassium bromide
- ☐ **D** sodium bromide

(b) Which of these compounds would be expected to have the **largest** difference between their experimental (Born–Haber) and theoretical lattice energies?

(1)

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- ☐ **A** calcium chloride
- ☐ **B** magnesium chloride
- ☐ **C** potassium bromide
- ☐ **D** sodium bromide

(Total for question = 2 marks)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The standard enthalpy change of solution of potassium chloride, KCl, is +17 kJ mol^{-1}.

The solubility of potassium chloride in water at 298 K is 359 g dm^{-3}.

Which of these explains the solubility of potassium chloride in water?

- ☐ **A** the hydration enthalpy of K^+ and the lattice energy of KCl are exothermic
- ☐ **B** the hydration enthalpy of K^+ and the lattice energy of KCl are endothermic
- ☐ **C** the total entropy change when KCl dissolves is positive
- ☐ **D** the total entropy change when KCl dissolves is negative

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(Total for question = 1 mark)

Q8.

* The table shows the theoretical and experimental (Born-Haber) lattice energy data for two metal halide

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compounds, sodium chloride and magnesium iodide.

Metal halide	Lattice energy / kJ mol^{-1}	
	Theoretical	Experimental (Born-Haber)
Sodium chloride	−770	−780
Magnesium iodide	−1944	−2327

Using the data, compare and contrast the type and strength of bonding in these compounds.
Give reasons for your answers.

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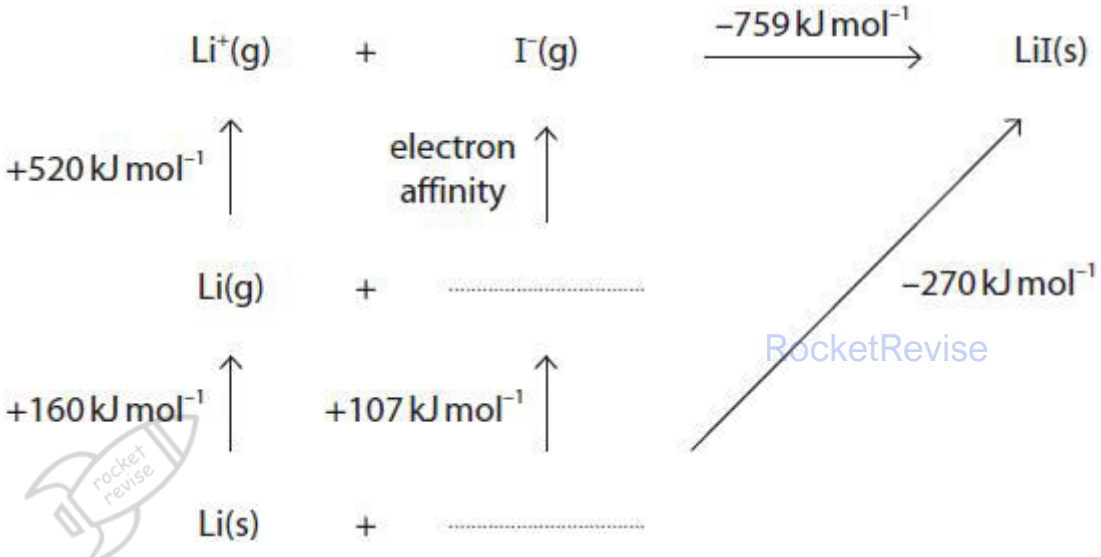
(6)

(Total for question = 6 marks)

Q9.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A simplified Born-Haber cycle for the formation of lithium iodide is shown.



(a) The enthalpy change of atomisation of iodine (+107 kJ mol^{-1}) is given by the equation

(1)

- ☐ **A** $\frac{1}{2}I_2(g) \rightarrow I(g)$
- ☐ **B** $\frac{1}{2}I_2(s) \rightarrow I(g)$
- ☐ **C** $I_2(g) \rightarrow 2I(g)$
- ☐ **D** $I_2(s) \rightarrow 2I(g)$

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(b) Use the information in the cycle to calculate the electron affinity of iodine.

(1)

- ☒ **A** –298 kJ mol^{-1}
- ☒ **B** –242 kJ mol^{-1}
- ☒ **C** +242 kJ mol^{-1}
- ☒ **D** +298 kJ mol^{-1}

(Total for question = 2 marks)

Q10.

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For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some energy changes are given in the table.

Energy change	Value / kJ mol^{-1}
Lattice energy for $CaCl_2(s)$	–2258
Enthalpy change of solution of $CaCl_2(s)$	–120
Enthalpy change of hydration of $Cl^-(g)$	–364

Use these data to calculate the enthalpy change of hydration of $Ca^{2+}(g)$.

- ☒ **A** –1410 kJ mol^{-1}
- ☒ **B** –1650 kJ mol^{-1}
- ☒ **C** –2014 kJ mol^{-1}
- ☒ **D** –3106 kJ mol–

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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these statements is **true**?

- ☒ **A** the enthalpy change of hydration of an ion is always negative
- ☒ **B** the enthalpy change of solution of an ionic compound is always negative

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☒ **C** an endothermic enthalpy change of solution indicates that an ionic compound will be insoluble

☒ **D** the more exothermic the lattice energy of an ionic compound, the more soluble it is likely to be

(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Experimental lattice energy values determined using the Born-Haber cycle often differ from calculated theoretical values.
The ionic radii and charges of some ions are shown.

P, Q, R and S are **not** the chemical symbols of the elements.

Ion	Ionic radius / nm
P^+	0.076
Q^+	0.167
R^-	0.133
S^-	0.220

Which of these compounds would be expected to show the largest percentage difference between the experimental and theoretical lattice energies?

☒ **A** **PR**
☒ **B** **PS**
☒ **C** **QR**
☒ **D** **QS**

(Total for question = 1 mark)

Q13.

This question is about oxides of iron.

(a) The equation for the reduction of iron(III) oxide by carbon is shown.

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$Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$

Some data relating to this reaction are shown.

Substance	Fe_2O_3	C	Fe	CO
$\Delta_f H^{\ominus}$ / kJ mol^{-1}	−824.2	0	0	−110.5
$S^{\ominus}$ / J K^{-1} mol^{-1}	87.4	5.7	27.3	197.6

(i) Calculate the standard enthalpy change, $\Delta_r H^{\ominus}$, for the reduction of iron(III) oxide by carbon.

(2)

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(ii) Calculate the entropy change, ΔS_{system}, for this reaction.

(3)

(iii) Calculate the temperature at which this reaction becomes feasible, using your answers from (a)(i) and (a)(ii).

(3)

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(b) The equation for the reduction of iron(III) oxide by carbon monoxide is shown.

$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g) \qquad \Delta H^{\ominus} = -24.8 \text{ kJ mol}^{-1}$

The entropy change, $\Delta S^{\ominus}_{system}$, for this reaction is +15.2 J mol^{-1} K^{-1}.

(i) Explain why the reaction should be feasible at any temperature.

(2)

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(ii) Explain how an increase in temperature would affect $\Delta S^{\ominus}_{total}$ total of this reaction. No calculation is required.

(3)

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(c) Iron also forms iron(II) oxide. The data in the table can be used to construct a Born-Haber cycle for iron(II) oxide, FeO.

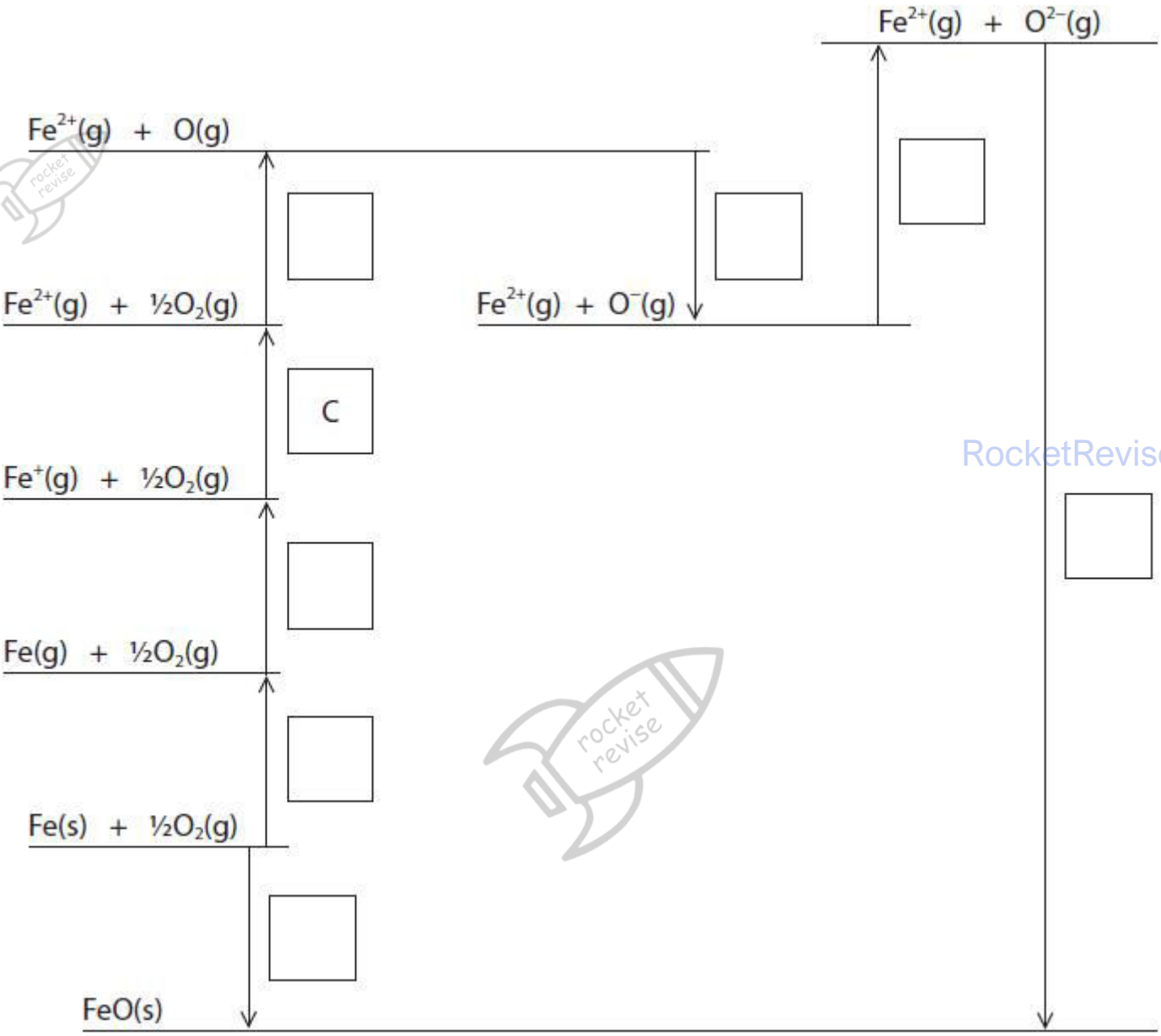
Letter	Enthalpy change	Value / kJ mol^{-1}
A	Enthalpy change of atomisation of iron	+416
B	First ionisation energy of iron	+759
C	Second ionisation energy of iron	
D	Enthalpy change of formation of iron(II) oxide	–272
E	First electron affinity of oxygen	–141
F	Second electron affinity of oxygen	+798
G	Lattice energy of iron(II) oxide	–3920
H	Enthalpy change of atomisation of oxygen ($\frac{1}{2}O_2(g) \rightarrow O(g)$)	+249

(i) Complete the Born-Haber cycle by putting letters in the boxes to label the energy changes.

(3)

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(ii) Calculate the value for second ionisation energy of iron, C.

(2)

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(iii) Suggest why the second electron affinity of oxygen is positive.

(2)

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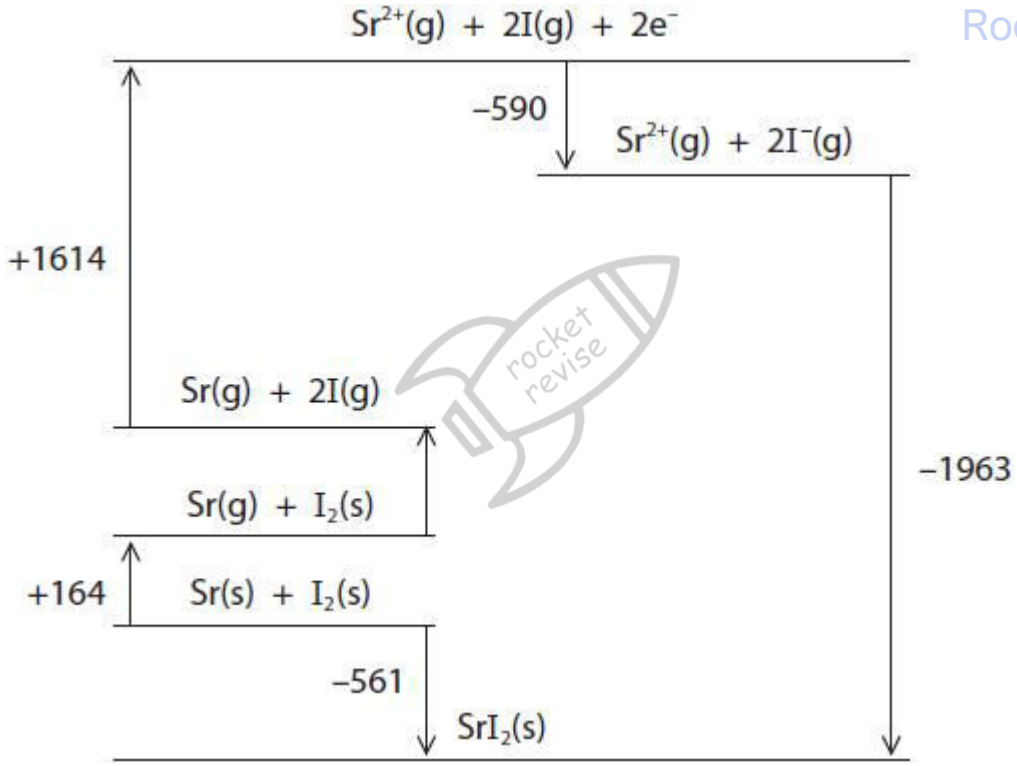
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(Total for question = 20 marks)

Q14.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The Born–Haber cycle for strontium iodide, SrI_2, is shown.
All values are in kJ mol^{-1}. The cycle is **not** to scale.



(a) Which of the following statements is **not** true?

(1)

- ☒ **A** the lattice energy of strontium iodide is exothermic
- ☒ **B** the first electron affinity of iodine is −295 kJ mol^{-1}
- ☒ **C** the second ionisation energy of strontium is +1614 kJ mol^{-1}
- ☒ **D** the standard enthalpy change of atomisation of strontium is +164 kJ mol^{-1}

(b) Using the Born–Haber cycle, it can be calculated that the standard enthalpy change of atomisation of iodine, in kJ mol^{-1}, is

(1)

- ☒ **A** +542
- ☒ **B** +271
- ☒ **C** +214
- ☒ **D** +107

(c) Theoretical lattice energies are calculated assuming only ionic bonding is present.

Which of these is most likely to be the **theoretical** lattice energy of strontium iodide?

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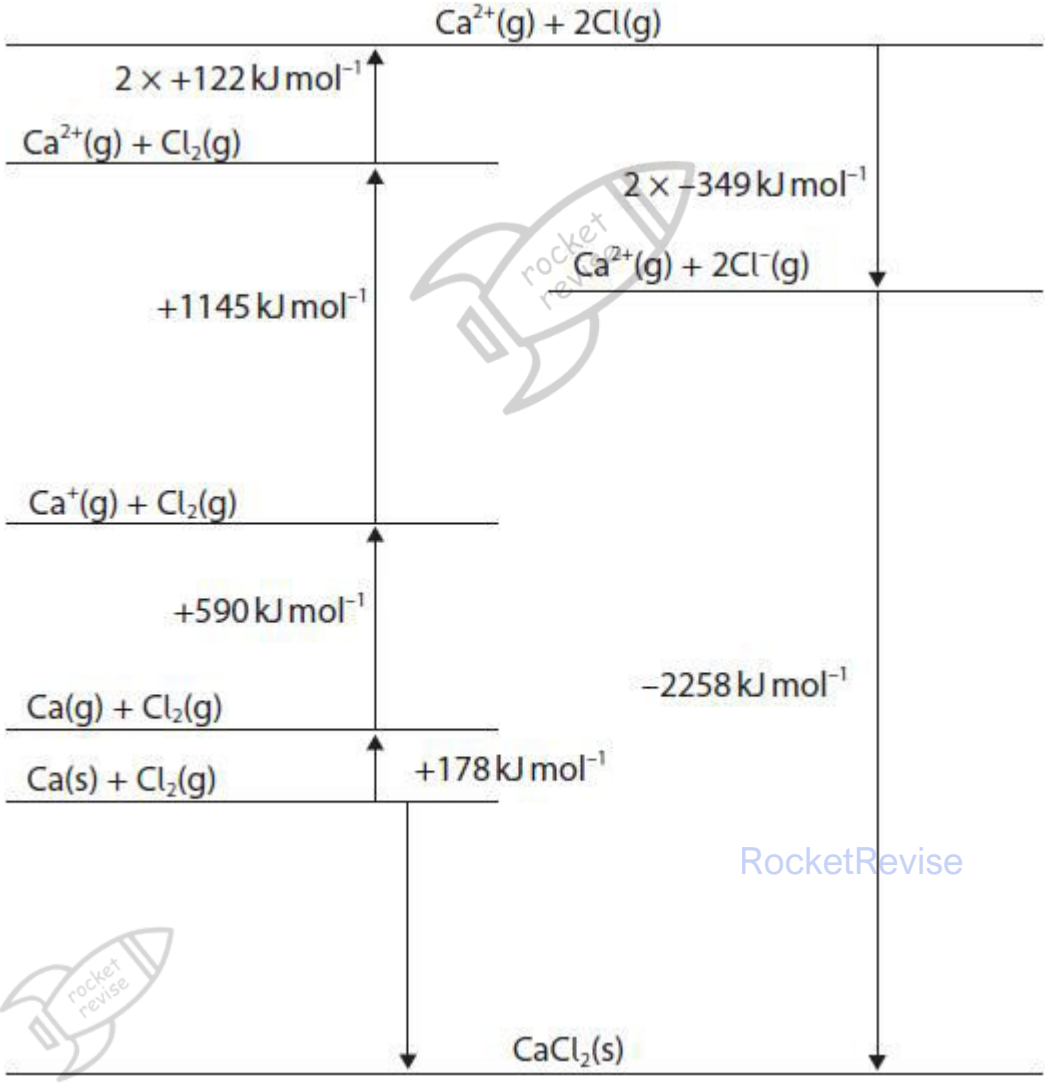
(1)

- ☒ **A** −561 kJ mol^{-1}
- ☒ **B** −1937 kJ mol^{-1}
- ☒ **C** −1963 kJ mol^{-1}
- ☒ **D** −1989 kJ mol^{-1}

(Total for question = 3 marks)

Q15.

A Born–Haber cycle for calcium chloride is shown.



(a) State the value of the $\Delta_{at}H$ for calcium.

(1)

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(b) Calculate the enthalpy change of formation for calcium chloride.

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(2)

(c) Some energy data are shown.

Compound	Theoretical lattice energy / kJ mol^{-1}	Experimental lattice energy / kJ mol^{-1}
$CaCl_2$	−2223	−2258
CaI_2	−1905	−2074

Explain why the difference between the theoretical and the experimental values for lattice energy is very much greater for calcium iodide than for calcium chloride.

(4)

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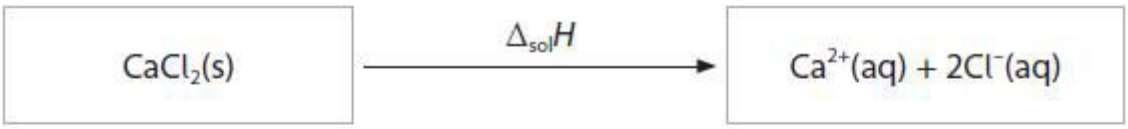
(d) Calcium chloride is soluble in water.

(i) Complete the energy cycle including labelled arrows.

(2)

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(ii) Calculate the enthalpy change of solution, $\Delta_{sol}H$, for calcium chloride using the data given and the completed energy cycle in (d)(i).

(2)

Data	Energy change / kJ mol^{-1}
LE ($CaCl_2(s)$)	−2258
$\Delta_{hyd}H$ ($Ca^{2+}(g)$)	−1579
$\Delta_{hyd}H$ ($Cl^-(g)$)	−378

(Total for question = 11 marks)

Q16.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An ionic compound is found to have some degree of covalent bonding.

What will be the cause of the covalency and how will the lattice energy obtained from a Born–Haber cycle differ from the theoretical value obtained from electrostatic theory?

		Cause of covalency	Born–Haber lattice energy compared with theoretical value
☐	**A**	cation polarised by anion	Born–Haber more exothermic
☐	**B**	anion polarised by cation	Born–Haber less exothermic
☐	**C**	cation polarised by anion	Born–Haber less exothermic
☐	**D**	anion polarised by cation	Born–Haber more exothermic

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(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q17.

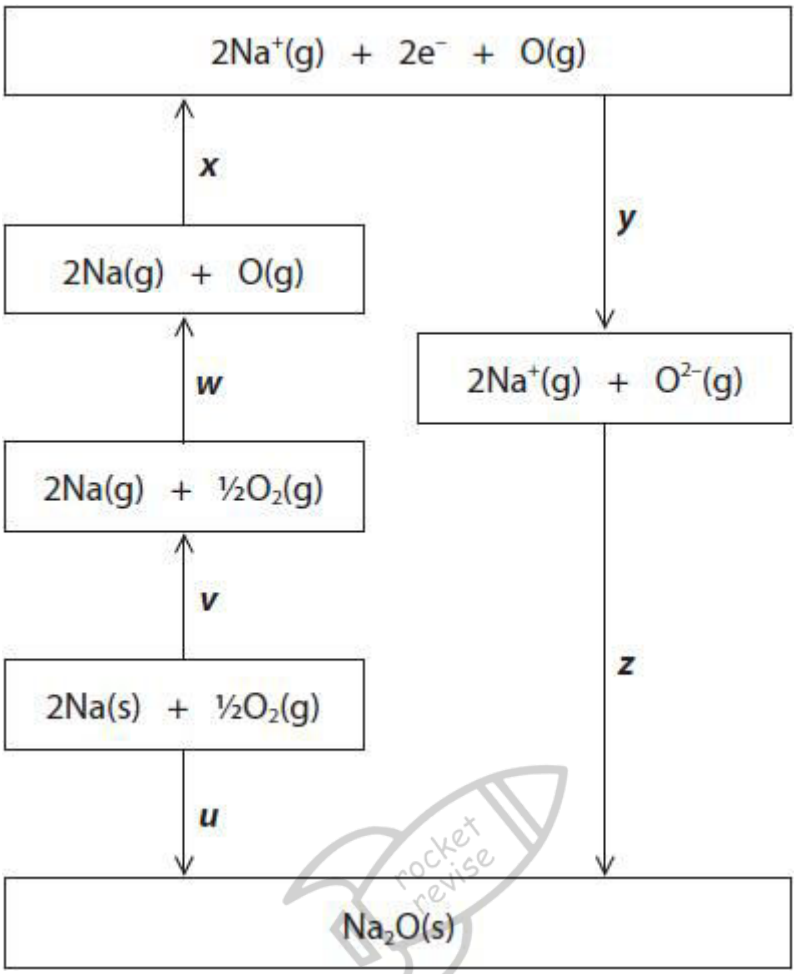
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the Born–Haber cycle for sodium oxide.

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(a) Which energy change in the cycle does ***y*** represent?

(1)

- ☒ **A** the first electron affinity of oxygen
- ☒ **B** the second electron affinity of oxygen
- ☒ **C** the sum of the first and second electron affinities of oxygen
- ☒ **D** the sum of the enthalpy change of atomisation of oxygen and the first and second electron affinities of oxygen

(b) Which expression gives the enthalpy change of formation of sodium oxide, $\Delta_f H$?

(1)

- ☒ **A** $\Delta_f H = \boldsymbol{v} + \boldsymbol{w} + \boldsymbol{x} - \boldsymbol{y} - \boldsymbol{z}$
- ☒ **B** $\Delta_f H = \boldsymbol{v} + \boldsymbol{w} + \boldsymbol{x} + \boldsymbol{y} - \boldsymbol{z}$
- ☒ **C** $\Delta_f H = \boldsymbol{v} + \boldsymbol{w} + \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{z}$
- ☒ **D** $\Delta_f H = \boldsymbol{v} + \boldsymbol{w} + \boldsymbol{x} + \boldsymbol{y} + \boldsymbol{z}$

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(Total for question = 2 marks)

Q18.

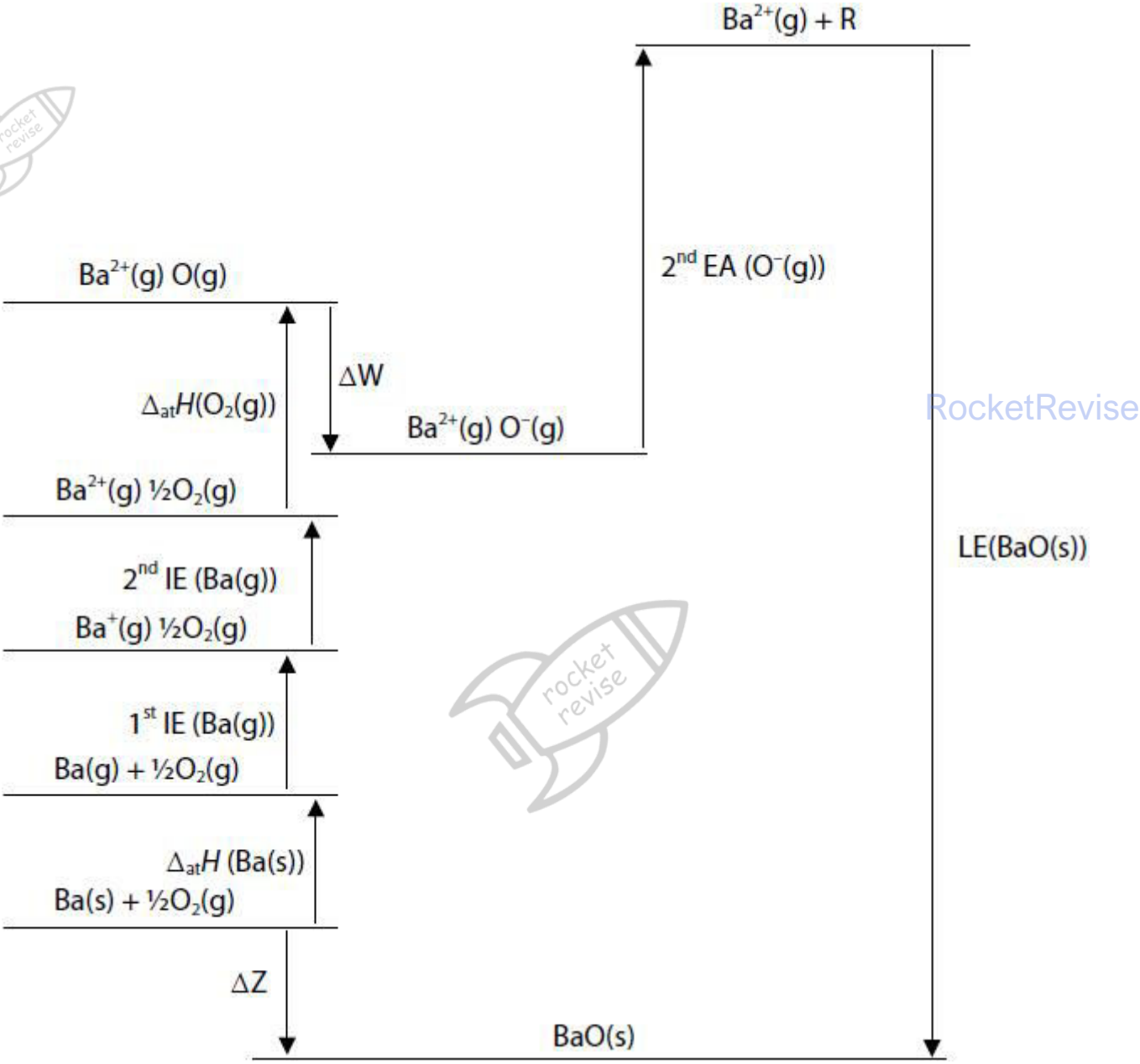
Barium oxide is an ionic compound that reacts with water to form barium hydroxide.

(a) A Born-Haber cycle for barium oxide, BaO, is shown.

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Some of the detail is missing. The letters **R**, **W** and **Z** represent some missing information.



(i) Identify the missing detail represented by the following letters.

(2)

R .....

W .....

(ii) Use the following data to calculate a value for the quantity ΔZ shown on the Born-Haber cycle. Include a sign and units in your answer.

(3)

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Energy quantity	Enthalpy change / kJ mol^{-1}
Enthalpy change of atomisation of barium, $\Delta_{at}H$(Ba(s))	+180.0
Enthalpy change of atomisation of oxygen, $\Delta_{at}H$ (½O_2(g))	+249.2
First ionisation energy of barium, Ist IE (Ba(g))	+503.0
Second ionisation energy of barium, 2nd IE (Ba(g))	+965.0
ΔW	−141.1
Second electron affinity of oxygen, 2nd EA (O^-(g))	+798.0
Lattice energy barium oxide, $\Delta_{LE}H$(BaO(s))	−3054.0

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(iii) The table gives some information about the lattice energies of barium oxide and magnesium iodide and shows the % difference between the theoretical and experimental values.

	Lattice energy / kJ mol^{-1}		
	Experimental	Theoretical	% difference
BaO(s)	−3054	−3029	0.8
MgI_2(s)	−2327	−1944	16.5

Explain why there is closer agreement for barium oxide than for magnesium iodide.

(3)

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(b) The table gives some information about Group 2 ions, M^{2+}, and their hydroxides.

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Formula of hydroxide	Lattice energy / kJ mol^{-1}	$\Delta_{sol}H$ / kJ mol^{-1}	Solubility / mol per 100 g	Ion	$\Delta_{hyd}HN^{2+}$ / kJ mol^{-1}
$Mg(OH)_2$	–3000	+150	2.0×10^{-5}	Mg^{2+}	–1930
$Ca(OH)_2$	–2640	+140	1.6×10^{-4}	Ca^{2+}	–1580
$Sr(OH)_2$	–2475	+105	3.3×10^{-4}	Sr^{2+}	–1450
$Ba(OH)_2$	–2230		2.4×10^{-4}	Ba^{2+}	–1360

(i) Calculate the enthalpy change of solution, $\Delta_{sol}H$, of $Ba(OH)_2$ using a fully-labelled Hess's cycle.
[The hydration enthalpy of the hydroxide ion, OH^- = –460 kJ mol^{-1}.]

(4)

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(ii) Explain why strontium hydroxide is slightly soluble in water, even though the enthalpy change of solution is endothermic.

(2)

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(Total for question = 14 marks)

Q19.

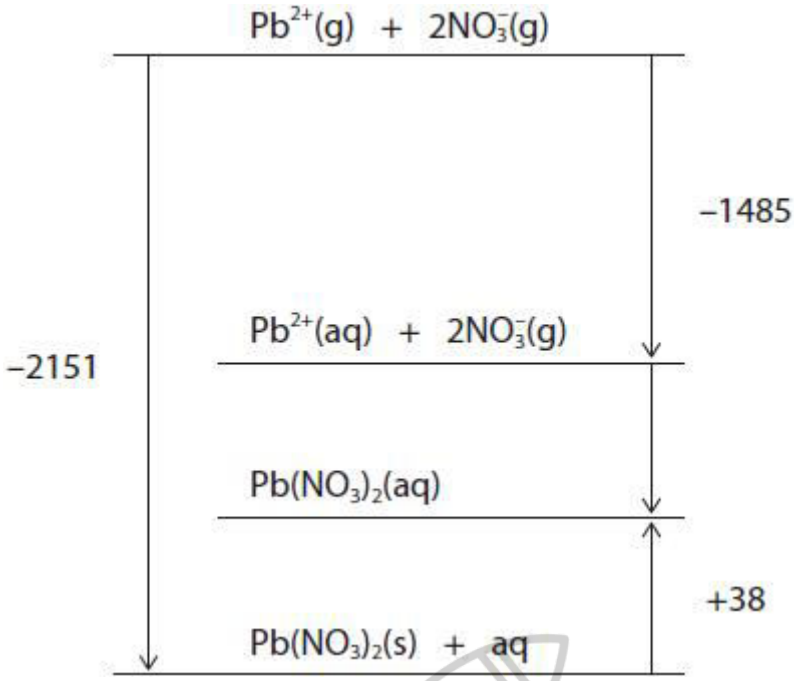
Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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An enthalpy cycle for lead(II) nitrate is shown.

All values are in kJ mol^{-1}. The cycle is **not** to scale.



(a) Which of these statements is **true**?

(1)

- ☒ **A** the enthalpy change of formation of lead(II) nitrate is −2151 kJ mol^{-1}
- ☒ **B** the enthalpy change of solution of lead(II) nitrate is exothermic
- ☒ **C** the enthalpy change of hydration of a lead(II) ion is more exothermic than that of a nitrate ion
- ☒ **D** the entropy change of the surroundings for dissolving lead(II) nitrate is positive

(b) Using the enthalpy cycle, it can be calculated that the enthalpy change of hydration of a nitrate ion, in kJ mol^{-1}, is

(1)

- ☒ **A** −3598
- ☒ **B** −1799
- ☒ **C** −628
- ☒ **D** −314

(Total for question = 2 marks)

Q20.

This question is about Group 2 compounds.

(a) Barium carbonate, $BaCO_3$, decomposes at high temperatures as shown.

$$BaCO_3(s) \rightarrow BaO(s) + CO_2(g)$$

(i) Calculate the standard entropy change for the system, $\Delta S^{\ominus}_{system}$.

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Compound	Standard molar entropy / $JK^{-1}\,mol^{-1}$
$BaCO_3$	112.1
BaO	70.4
CO_2	213.6

(2)

(ii) The lowest temperature at which barium carbonate, $BaCO_3$, decomposes is 712 °C.
Use this information and your answer from (a)(i) to calculate the standard enthalpy change of the reaction, in kJ mol^{-1} .
Give your answer to an appropriate number of significant figures.

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(3)

(b) Magnesium chloride, $MgCl_2(s)$, is used in the manufacture of tofu from soya milk.

An experiment was carried out to determine the enthalpy change of solution, $\Delta_{sol}H$, of anhydrous magnesium chloride, $MgCl_2(s)$.
4.26 g of anhydrous magnesium chloride was added to 200 cm^3 of deionised water in a polystyrene cup. The mixture was stirred to ensure all the solid dissolved to form a solution.
The temperature of the solution rose by 6.8 °C.
Calculate the enthalpy change of solution, $\Delta_{sol}H$, of anhydrous magnesium chloride, $MgCl_2(s)$.
Include a sign and units with your answer.
[Specific heat capacity of the solution = 4.18 J g^{-1} $°C^{-1}$]

(3)

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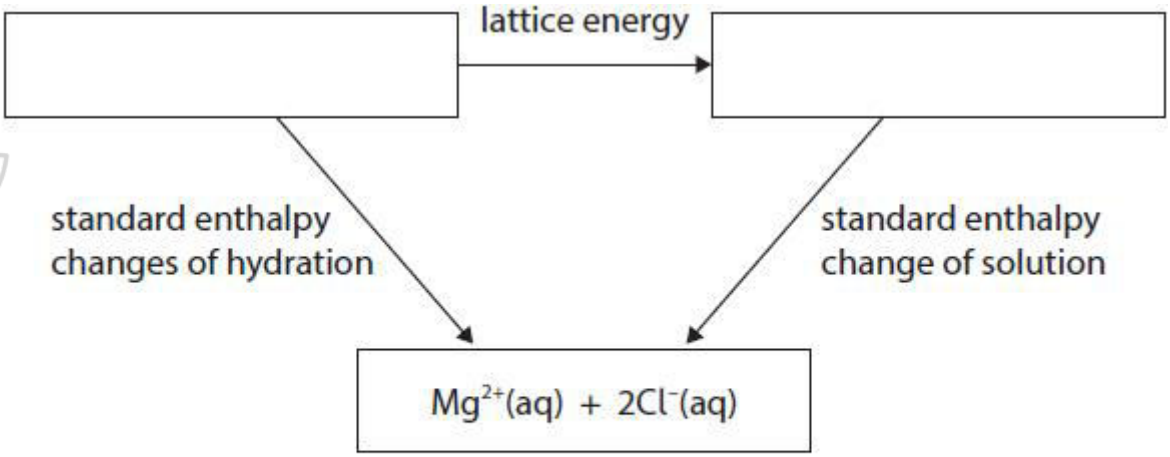
(c) The standard enthalpy change of solution, $\Delta_{sol}H^{\ominus}$, of anhydrous $MgCl_2(s)$, can also be determined using a Hess Cycle.

(i) Complete the Hess Cycle shown by adding appropriate formulae, with state symbols, to the empty boxes.

(1)

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[Lattice energy = −2526 kJ mol^{-1}
Standard enthalpy change of hydration of magnesium ions = −1920 kJ mol^{-1}]

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(ii) Calculate the standard enthalpy change of hydration of chloride ions, $\Delta_{hyd}H^{\ominus}[Cl^-(g)]$, using the cycle and data in (c)(i) and your answer to (b).

[If you did not obtain a final answer to (b) use a value of −155 kJ mol^{-1} . This is not the correct value.]

(3)

(d) The theoretical lattice energy for magnesium chloride, $MgCl_2(s)$, is −2326 kJ mol^{-1}.

(i) Give **two** assumptions used in theoretical calculations of lattice energy.

(2)

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(ii) Explain the difference between the theoretical and experimental values for the lattice energy of magnesium chloride, $MgCl_2(s)$.

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(3)

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(e) Hydrated barium chloride is a soluble, toxic salt. It can be formed from the reaction of barium carbonate with hydrochloric acid as shown.

$$BaCO_3 + 2HCl + nH_2O \rightarrow BaCl_2 \bullet nH2O + CO_2 + H_2O$$

A 5.00 g sample of barium carbonate reacted with 120 cm^3 of 0.500 mol dm^{-3} hydrochloric acid.

(i) Show that the hydrochloric acid is in excess in this reaction.

(2)

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(ii) All of the barium carbonate reacted and produced 6.19 g of hydrated barium chloride. Calculate the relative formula mass of $BaCl_2 \bullet nH_2O$ and hence deduce the value of n.

(3)

(Total for question = 22 marks)

Q21.

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The question is about lattice energies.

The table shows energy values used in a Born–Haber cycle for magnesium chloride, $MgCl_2$.

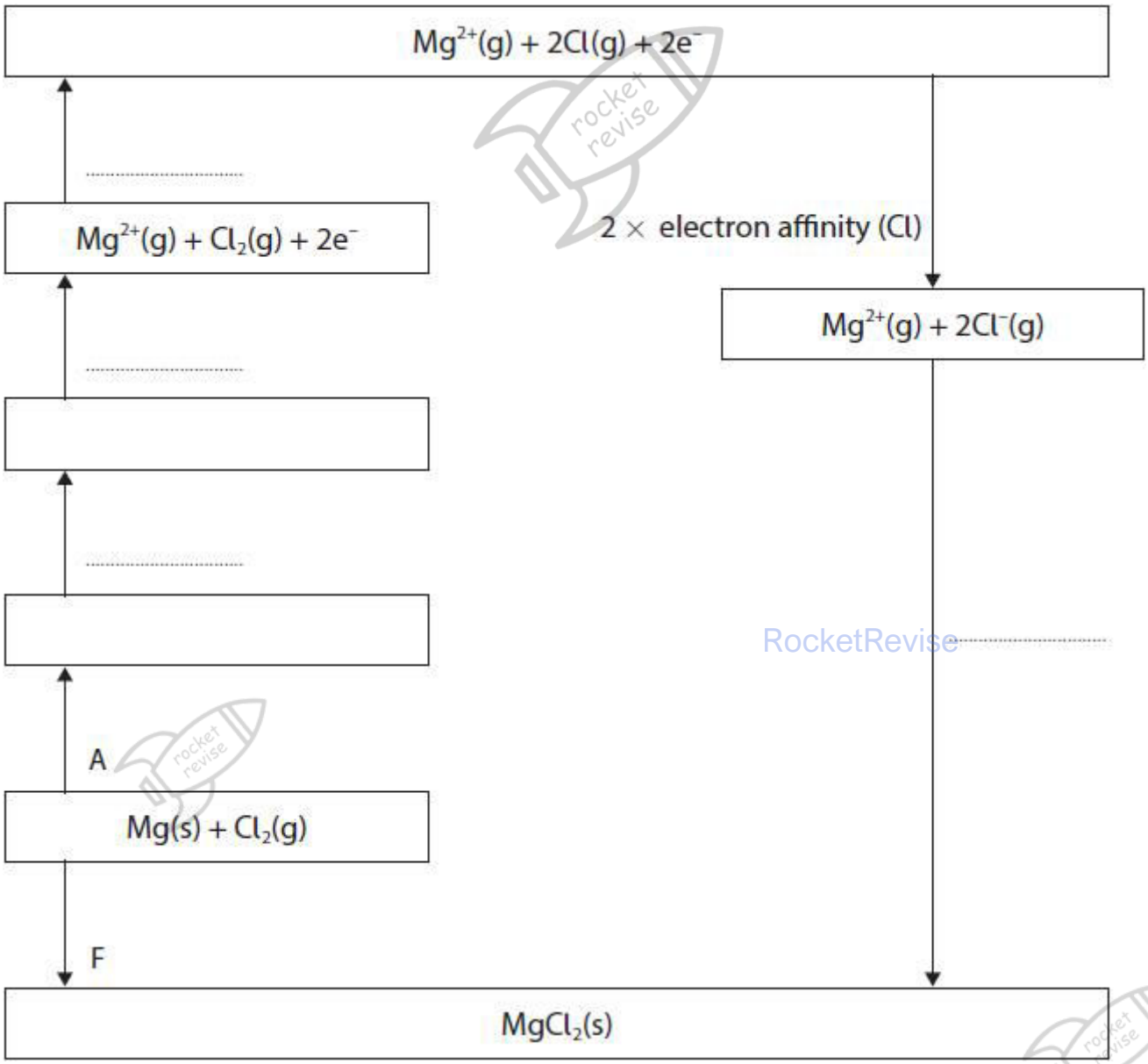
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Energy change	Label	Value / kJ mol^{-1}
Enthalpy change of atomisation of magnesium	A	+148
First ionisation energy of magnesium	B	+738
Second ionisation energy of magnesium	C	+1451
Enthalpy change of atomisation of chlorine	D	+122
Lattice energy of magnesium chloride	E	–2526
Enthalpy change of formation of magnesium chloride	F	–641

(a) (i) Complete the Born–Haber cycle for magnesium chloride by adding labels for each of the four energy changes and writing formulae in the two empty boxes.

(3)



(ii) Calculate a value for the electron affinity of chlorine, in kJ mol^{-1}, using the data in the table and the

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completed Born–Haber cycle.

(2)

(iii) Explain why, when magnesium reacts with chlorine, $MgCl_2$ is formed rather than $MgCl_3$.

(2)

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(iv) Calculate the standard molar enthalpy change of solution of magnesium chloride, in kJ mol^{-1}, using the data shown and the value for the lattice energy, LE[$MgCl_2$], given in the table.

Data $\Delta_{hyd}H^{\ominus}[Mg^{2+}(g)] = -1920\,kJ\,mol^{-1}$ $\Delta_{hyd}H^{\ominus}[Cl^{-}(g)] = -364\,kJ\,mol^{-1}$

(2)

*(b) Lattice energies from the Born–Haber cycle are based on experimental values. Theoretical lattice energies can also be calculated. Experimental and theoretical values for three different crystal lattices are shown.

Compound	Experimental lattice energy / kJ mol^{-1}	Theoretical lattice energy / kJ mol^{-1}
sodium fluoride NaF	–918	–912
magnesium fluoride MgF_2	–2957	–2913
magnesium chloride $MgCl_2$	–2526	–2326

Discuss the reasons for the differences in these six values of lattice energy in terms of the structure and bonding in these three substances.

(6)

(Total for question = 15 marks)

Q22.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Some data on lattice energies are shown.

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Compound	Experimental lattice energy / kJ mol^{-1}	Theoretical lattice energy / kJ mol^{-1}
NaI	−705	−687
AgBr	−891	−816
Rb_2S	−1944	−1904
Li_2O	−2814	−2799

Which compound has the greatest degree of covalency within its ionic bonds?

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- ☒ **A** NaI
- ☒ **B** AgBr
- ☒ **C** Rb_2S
- ☒ **D** Li_2O

(Total for question = 1 mark)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the standard enthalpy change of atomisation of iodine?

- ☒ **A** $\frac{1}{2}I_2(s) \rightarrow I(g)$
- ☒ **B** $\frac{1}{2}I_2(g) \rightarrow I(g)$
- ☒ **C** $I_2(s) \rightarrow 2I(g)$
- ☒ **D** $I_2(g) \rightarrow 2I(g)$

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(Total for question = 1 mark)

Q24.

Which equation represents the second electron affinity of oxygen?

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☐ **A** $O^-(g) + e^- \rightarrow O^{2-}(g)$

☐ **B** $O(g) + 2e^- \rightarrow O^{2-}(g)$

☐ **C** $O^+(g) \rightarrow O^{2+}(g) + e^-$

☐ **D** $O(g) \rightarrow O^{2+}(g) + 2e^-$

(Total for question = 1 mark)

Q25.

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Many effervescent products such as vitamin tablets are solids which contain citric acid, $C_6H_8O_7$, and sodium hydrogencarbonate. Only when these tablets are added to water is fizzing observed. The equation for the reaction is shown.

$$C_6H_8O_7 + 3NaHCO_3 \rightarrow 3H_2O + 3CO_2 + Na_3C_6H_5O_7$$

(a) Give **two** reasons why this reaction is thermodynamically feasible when considering the entropy of the system.

(2)

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(b) Describe, in terms of the particles involved, why the reaction does not occur until the tablets are dissolved in water.

(3)

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(c) For the reaction at 25°C, the standard entropy change of the surroundings, $\Delta S^{\ominus}_{surroundings} = -234.9\,JK^{-1}\,mol^{-1}$.

Calculate the enthalpy change of the reaction.
Include a sign and units in your answer.

(2)

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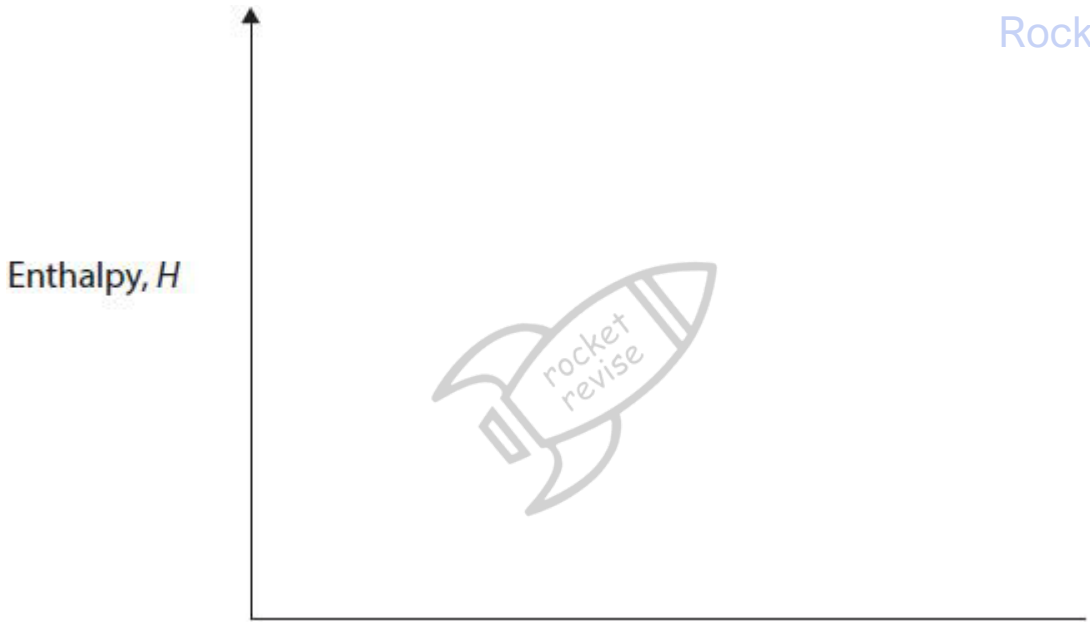
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(d) The enthalpy change of solution, $\Delta_{sol}H$, of sodium hydrogencarbonate is +18.6 kJ mol^{-1}.

Sketch the enthalpy level diagram to show the relationship between the enthalpy change of solution, lattice energy and enthalpies of hydration for the dissolving of sodium hydrogencarbonate in water. Fully label your diagram.

(3)

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(Total for question = 10 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound has the greatest covalent character?

- ☐ **A** $CaBr_2$
- ☐ **B** CaI_2
- ☐ **C** $MgBr_2$
- ☐ **D** MgI_2

(Total for question = 1 mark)

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Q27.

This question is about some copper compounds.

(a) Crystals of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, lose water when heated, forming anhydrous copper(II) sulfate, $CuSO_4$.

$$CuSO_4 \cdot 5H_2O(s) \rightarrow CuSO_4(s) + 5H_2O(l)$$

Substance	$CuSO_4 \cdot 5H_2O(s)$	$CuSO_4(s)$	$H_2O(l)$
Standard enthalpy of formation, $\Delta_f H^\ominus$ / kJ mol^{-1}	−2279.6	−771.4	−285.8
Standard molar entropy, $S^\ominus$ / J K^{-1} mol^{-1}	300.4	109.0	69.9

(i) Using the data in the table, calculate the standard enthalpy change, $\Delta_r H^\ominus$, for this reaction.

(2)

(ii) Using the data in the table, calculate the standard molar entropy change, $\Delta S^\ominus_{system}$, for this reaction.

(2)

(iii) Using your answers from (a)(i) and (a)(ii), calculate the **lowest** temperature at which hydrated copper(II) sulfate will decompose.

(3)

(b) Hydrated copper(II) sulfate is soluble in water. A student carried out an experiment to determine its enthalpy change of solution.

100 g of distilled water was placed in a polystyrene cup and the temperature recorded.
21.36 g of hydrated copper sulfate was added to the distilled water, the mixture stirred, and the minimum temperature recorded.
Using this data, the student's calculated enthalpy change of solution was +12.3 kJ mol^{-1}.
Calculate the temperature change in this experiment.
[specific heat capacity of the solution is 3.7 J g^{-1} °C^{-1}
total mass of solution = 121.36 g
molar mass of $CuSO_4 \cdot 5H_2O$ = 249.6 g mol^{-1}]

(4)

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(c) Copper(II) chloride, $CuCl_2$, is another copper compound that is soluble in water. It is possible to calculate the enthalpy change of solution, $\Delta_{sol}H$, using a Hess cycle and the lattice energy and enthalpy changes of hydration, $\Delta_{hyd}H$.

(i) Explain what is meant by the term **enthalpy change of hydration**.

(2)

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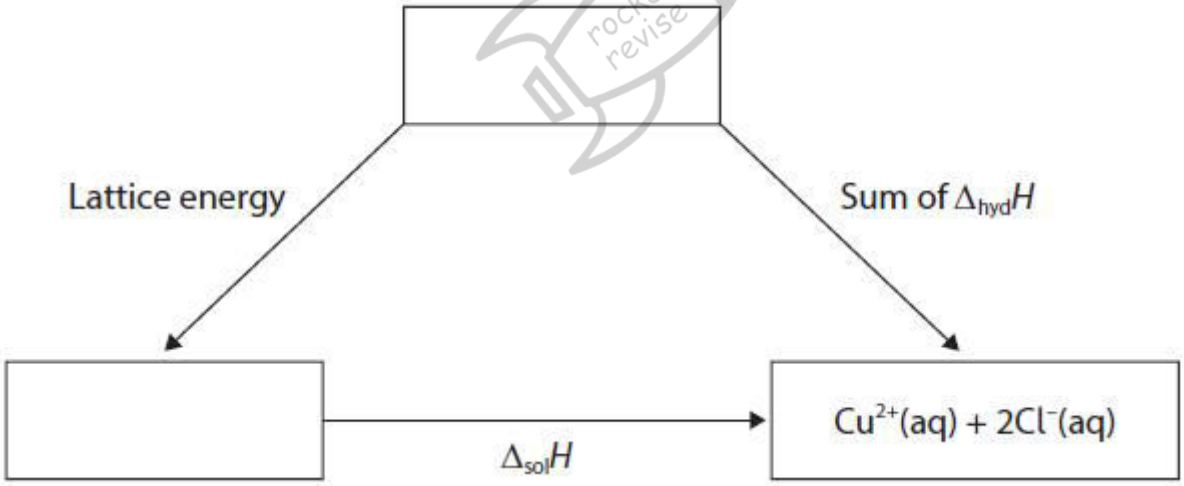
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(ii) Complete the Hess cycle by filling in the empty boxes.

(1)



(iii) Complete the expression for the enthalpy change of solution using the enthalpy changes of hydration and lattice energy.

$\Delta_{sol}H$ =

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(1)

(iv) The lattice energy of copper(II) chloride is −2811 kJ mol^{-1}.
Use this value, the data in the table, and your expression in (c)(iii) to calculate the enthalpy change of solution.

(2)

Ion	Enthalpy change of hydration / kJ mol^{-1}
Cu^{2+}	−2100
Cl^{-}	−378

(v) The enthalpy change of hydration of Cu^{+} is −593 kJ mol^{-1}.

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Explain why the hydration enthalpy of Cu^+ is much less exothermic than Cu^{2+}.

(2)

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(Total for question = 19 marks)

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